

SAT Math Mastery · Week 1A

Linear Equations: Structure, Rearranging & Translating

Time 120 minutes: 90 min solving + 30 min self-check and error log

Topics one-variable equations · solution-count structure · identities & coefficient matching · literal equations · word translation · inequalities

Name / Date _____

1 · SKILLS & IDEAS

- S1.** Solve any linear equation (distribute, LCD for fractions/decimals, collect, divide).
- S2.** Spot a repeated *block* and solve for the block instead of the variable.
- S3.** Decide *one* / *none* / *infinitely many* solutions by comparing coefficients, before solving.
- S4.** Match coefficients when an equation is true for every x ; solve for constants.
- S5.** Rearrange literal equations (make any letter the subject).
- S6.** Translate words: “less than”, consecutive integers, ratios, fixed-plus-rate, ages.
- S7.** Linear inequalities: sign flip, compound statements, integer counts, rounding to a feasible whole number.
- S8.** Absolute-value equations: split into two cases and test each answer.

2 · REQUIRED FACTS

- Equal operations on both sides keep an equation true; multiplying or dividing an inequality by a negative number reverses it.
- “ n less than m ” = $m - n$; “at least” $\geq$; “at most” / “no more than” $\leq$; “more than” $>$.
- Consecutive integers differ by 1; consecutive *even or odd* integers differ by 2.
- A ratio $a : b = m : n$ means $a = mt$, $b = nt$; the whole is $(m + n)t$.
- $|u| = v$ means $u = v$ or $u = -v$, and requires $v \geq 0$.
- A sequence with a constant difference d has n th term $a_1 + (n - 1)d$.

3 · REQUIRED FORMULAS

$$ax + b = cx + d \implies (a - c)x = d - b, \quad x = \frac{d - b}{a - c} \quad (a \neq c)$$

$$P_{\text{rect}} = 2(\ell + w), \quad \text{total} = \text{fixed fee} + \text{rate} \times \text{quantity}$$

$$n, n + 2, n + 4, n + 6 \quad (\text{consecutive odd}), \quad \text{sum of } k \text{ terms} = k \cdot (\text{mean})$$

4 · MINI THEORY SHEET

Master reduction. Every linear equation collapses to $mx = n$. Then:

$m \neq 0$	one solution	$x = \frac{n}{m}$
$m = 0, n \neq 0$	no solution	$0 = n$ is false
$m = 0, n = 0$	infinitely many	$0 = 0$ is true for all x

Identity in x . If two expressions are equal for every x , the coefficients of x match *and* the constants match.

Block rule. If an expression E occurs, $cE = k \implies E = \frac{k}{c}$; scale or shift the block, never expand it.

Inequalities. Same steps as equations; reverse only on $\times/\div$ by a negative. Compound: operate on all three parts.

Contexts. Fixed + rate: $y = \text{fee} + \text{rate} \cdot n$. For “at most” counts solve, then round down; for “at least” counts round up.

Absolute value. Solve $u = v$ and $u = -v$, then substitute each answer into the original equation.

5 · GUIDED EXAMPLES

Example 1 — Structure first (parameter)

For what value of k does $4(x - 2) + k = 4x - 3$ have infinitely many solutions?

Expand: $4x - 8 + k = 4x - 3$. The x -coefficients agree, so the constants must agree: $k - 8 = -3$, so $k = 5$.

Example 2 — Fractions: clear with the LCD

Solve $\frac{x+1}{2} - \frac{x-3}{5} = 2$.

LCD = 10: $5(x+1) - 2(x-3) = 20 \Rightarrow 3x + 11 = 20 \Rightarrow x = 3$.

Example 3 — Words $\rightarrow$ equation (consecutive integers)

The sum of three consecutive even integers is 54. What is the largest?

Let them be n , $n + 2$, $n + 4$: $3n + 6 = 54 \Rightarrow n = 16$. The integers are 16, 18, 20; the largest is 20.

6 · PRACTICE SET — 25 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

EASY · NO CALCULATOR NEEDED

If $f(x) = 6(2x + 4)$, for what value of x is $f(x) = 48$?

(A) 8

(B) 2

(C) 4

(D) 6

2.

EASY · NO CALCULATOR NEEDED

What value of x satisfies the equation $\frac{2x + 5}{3} = x - 4$?

(A) -7

(B) 7

(C) 17

(D) 27

3.

EASY · NO CALCULATOR NEEDED

The inequality $-5(x - 1) \leq 3(x + 3)$ is equivalent to which of the following?

(A) $x \geq -\frac{1}{2}$ (B) $x \geq \frac{1}{2}$ (C) $x \leq \frac{1}{2}$ (D) $x \leq -\frac{1}{2}$

4.

EASY · NO CALCULATOR NEEDED

If $p = \frac{5q-3}{6}$, which expression is equal to q ?

(A) $\frac{p}{5} + 3$

(B) $\frac{6p-3}{5}$

(C) $\frac{5p+3}{6}$

(D) $\frac{6p+3}{5}$

5.

EASY · NO CALCULATOR NEEDED

$7x + 28 = 7x + k$. The equation has infinitely many solutions. What is the value of k ?

6.

EASY · NO CALCULATOR NEEDED

If $58 = 22 + 3(x - 5)$, what is the value of $3(x - 5)$?

7.

MEDIUM · NO CALCULATOR NEEDED

m , n , and p are three consecutive positive odd integers with $m < n < p$. Their sum is 63. What is the value of mp ?

8.

MEDIUM · NO CALCULATOR NEEDED

What is the solution to $\frac{3x+5}{4} = \frac{4x-1}{5}$?

9.

MEDIUM · NO CALCULATOR NEEDED

If $\frac{3m}{4} + 2 = 8$, what is the value of $\frac{5}{2m}$?

10.

MEDIUM · NO CALCULATOR NEEDED

A woodworker charges a one-time fee of \$45 plus \$17 for each hour of work. The total charge for one job was \$164. For how many hours did the woodworker work?

- (A) 119 (B) 5
(C) 7 (D) 9

11.

MEDIUM · CALCULATOR

A van can carry at most 3,600 kg. It is already loaded with 40 boxes of 25 kg each. Each additional crate weighs 18 kg. What is the greatest number of additional crates that can be loaded?

12.

MEDIUM · NO CALCULATOR NEEDED

Let a and b be numbers such that $20 < a < 30$ and $45 < b < 60$. Which of the following represents all possible values of $a - b$?

(A) $-15 < a - b < 40$

(B) $-40 < a - b < -15$

(C) $15 < a - b < 40$

(D) $-40 < a - b < 15$

13.

MEDIUM · NO CALCULATOR NEEDED

The solution to $3x - k = 87$ is $x = 174$, where k is a constant. What is the value of k ?

14.

MEDIUM · NO CALCULATOR NEEDED

$N = \frac{Sp}{q-1}$, where N , S , and p are nonzero. Which expression is equal to q ?

(A) $\frac{Sp+1}{N}$

(B) $\frac{Sp}{N} - 1$

(C) $\frac{N}{Sp} + 1$

(D) $\frac{Sp}{N} + 1$

15.

MEDIUM · NO CALCULATOR NEEDED

Daniel is now five times as old as his sister Emma. In 4 years, Daniel will be three times as old as Emma will be then. How old is Daniel now?

16.

HARD · NO CALCULATOR NEEDED

$a(7x - 14) + 5a = 7(ax - 2a) - 35$. For what value of a does the equation have infinitely many solutions?

17.

HARD · NO CALCULATOR NEEDED

If $a(x - 3) + 5 = bx - 2a + 11$ is true for all values of x , where a and b are constants, what is the value of a ?

18.

HARD · NO CALCULATOR NEEDED

$$\frac{18x + 24}{6} - \frac{s}{11} = r(x - 9)$$

In the equation above, s and r are constants and $s > 0$. If the equation has infinitely many solutions, what is the value of s ?

19.

HARD · NO CALCULATOR NEEDED

Let m and n be positive integers such that one-third of m is n less than one-half of m . Which of the following could be the value of m ?

(A) 12

(B) 16

(C) 8

(D) 10

20.

HARD · NO CALCULATOR NEEDED

For a certain number c , a sequence begins c , $2c + 3$, $3c + 6$, $4c + 9$, ... and the pattern continues. If the fifth term is 52, what is the tenth term?

21.

HARD · NO CALCULATOR NEEDED

The smallest integer in a set of consecutive integers is -11 . The sum of the integers in the set is 12. How many integers are in the set?

22.

HARD · NO CALCULATOR NEEDED

Maya is 4 years older than Noah. The square of Maya's age, in years, is 64 greater than the square of Noah's age. What is the sum, in years, of their ages?

23.

HARD · DESMOS

What is the value of x that satisfies $|3x + 2| = 5x - 6$?

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

For each integer $k \neq 1$, the equation $(k + 1)x + 3 = 2x + k$ has exactly one solution x . What is the sum of all integer values of k for which x is also an integer?

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the triangle shown (not drawn to scale), the side lengths are $x + 2$, $3x - 2$, and $5x - 16$. The triangle is isosceles. What is the sum of all possible perimeters?

